Experiment 2 — Hoisting Mechanism

Design of Mechanisms (BE05000341) · Quiz Solutions

Q1. Why are forged hooks preferred over cast hooks?

A crane hook lifts heavy, often shock-type loads and its failure can injure people, so it must be made by a process that gives the maximum strength, reliability and a visible warning before failure. Forging is chosen over casting for the following main reasons:

- **Grain flow follows the shape.** During forging, the metal grains are squeezed and aligned along the shape of the hook, which gives high strength and toughness exactly where the stress acts (at the inner fibre of the trapezoidal cross-section where tensile stress is maximum).
- **No internal defects.** Forging closes up blow-holes, porosity and shrinkage cavities that are common in castings and act as hidden crack-starters under cyclic load.
- **Better fatigue and impact resistance.** A hook is loaded and unloaded thousands of times (every lift); forged steels such as **30C8** or **40C8** handle this far better than cast iron or cast steel.
- **Ductile failure gives warning.** A forged hook bends and stretches visibly before it breaks, so the operator can take corrective action; a cast hook is more brittle and may fail suddenly without any sign.
- **Uniform material and reliable design.** Properties are consistent throughout the hook, so the designer can trust the permissible stress used in the calculations and the actual factor of safety stays close to the design value.
- **Better surface finish and easier inspection.** The polished surface after forging and machining is easy to inspect by magnetic-particle or dye-penetrant testing — a cast surface hides small defects.

Q2. What is the purpose of a sheave?

A **sheave** is a grooved pulley over which the wire rope runs. It is a basic element of every hoist and crane and serves several purposes at the same time:

- **Guides** the rope along a definite path and **changes its direction of pull** (for example, from the drum down to the hook).
- **Provides mechanical advantage** when used as a movable sheave in a block-and-tackle, so a smaller effort P can lift a larger load W ($W = n \times P \times \eta$, where n = number of rope parts). The trade-off is that the effort has to travel n times the distance.
- **Reduces friction**, because the rope rolls over the sheave on its own bearing instead of sliding over a fixed surface. This cuts the rope wear and the power lost.
- **Increases rope life** — a large groove diameter (typically **D = 20 – 40 × rope dia**) keeps the bending stress in the individual wires low, and the semicircular groove supports the rope evenly without flattening it.

- **Keeps the rope on the drum straight**, by acting as a guide-pulley above the drum so the fleet angle stays within the recommended ± 4°.

Q3. What factor of safety is used in lifting equipment?

Lifting equipment carries heavy loads, is subject to shocks, and failure directly endangers life. For this reason the factor of safety (FOS) is kept much higher than in general machine design. The exact value depends on the component and on the applicable code (IS 2266 for wire ropes, IS 807 for cranes, IS 2938 for lifts).

Component	Typical FOS	Remark
Wire rope for cranes and hoists	5 – 6	Based on the rope's breaking load
Wire rope for passenger lifts / mine hoists	8 – 12	Higher because lives are at risk
Forged crane hook	4 – 6	Based on yield or ultimate strength
Lifting chains	4 – 5	Depends on the chain grade
Pins, shafts and load-carrying parts	3 – 4 or more	Higher under shock loading

For a specific problem, always **use the FOS stated in the question** — for example FOS = 5 in Q8 and FOS = 30 in Q9 of this experiment.

Q4. Construction of a 6 × 19 wire rope.

A steel wire rope is built up from very thin, high-carbon-steel wires arranged in several twisted layers so that it is both strong and flexible. The designation "6 × 19" describes exactly how the wires are grouped:

- **6 strands** are wound in a helix around a central **core** (so the "6" in 6 × 19 is the number of outer strands you see on the outside of the rope).
- **Each strand has 19 steel wires** laid out in three layers: 1 centre wire + 9 inner wires + 9 outer wires (sometimes written as 1 + 9 + 9).
- The **core** is of two common types: **Fibre Core (FC)** — natural or synthetic fibre; soft, holds lubricant and gives flexibility — or **Independent Wire Rope Core (IWRC)** — a small 7 × 7 wire rope used as a core; stronger, used where crushing or heat is a problem.
- **Right-hand (RH) ordinary lay** is the most common: outer strands are twisted right-handed, and the wires within each strand are twisted the opposite way, which stops the rope from twisting under load.
- The high-tensile wires are usually of **1570 N/mm² or 1770 N/mm²** grade, often **galvanised** to resist corrosion. 6 × 19 is the most common general-purpose crane and hoist rope — flexible enough for small sheaves and still carrying high load.

Q5. Designation of a steel wire rope.

A rope is written in a standard order so that any maker, inspector or user understands exactly what is being specified:

nominal diameter (mm) × construction × core × tensile grade × lay × finish × standard

A typical example for a general crane rope:

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18 mm · 6 × 19 · FC · 1770 N/mm² · Right-hand ordinary lay · galvanised · IS 2266
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Meaning of each item: **18 mm** = nominal rope diameter; **6 × 19** = construction (6 strands of 19 wires); **FC** = Fibre Core; **1770 N/mm²** = ultimate tensile strength of the wires; **RH ordinary lay** = strands twisted right-hand with wires in each strand twisted the opposite way; **galvanised** = zinc-coated for corrosion protection; **IS 2266** = Indian Standard the rope is made and tested to.

Q6. Guidelines for the selection of wire ropes.

Picking a wire rope is not just about the load — several things decide the final choice. The common guidelines, taken from IS 2266, IS 2365 and the equipment code of practice, are:

- **Load class and shock.** Decide whether the duty is light, medium, heavy or very heavy and apply the appropriate **factor of safety** (5 – 12 as shown in Q3).
- **Sheave and drum diameter.** Keep **D / d ≥ 20 – 40** so the wires do not bend too sharply; a bigger D/d means fewer bending stress cycles and a longer rope life. Mine and lift ropes use the higher values.
- **Operating environment.** For moisture or corrosive air use **galvanised** rope; for hot or crushing conditions use **IWRC**; for abrasive dust use larger-diameter outer wires (6 × 7 or 6 × 19 Seale).
- **Flexibility needed.** More wires per strand (6 × 37 or 8 × 19) → more flexible, suited to small sheaves and frequent bending; fewer wires (6 × 7) → stiffer, more wear-resistant.
- **Lay direction and type.** Ordinary (regular) lay resists kinking and is the usual choice; Lang's lay has lower bending stress and longer life but can rotate under free load, so it is used only where both ends are fixed.
- **Code of practice.** Follow **IS 2266 / IS 2365** for the rope itself and the relevant equipment code (IS 807 for cranes, IS 2938 for lifts).
- **Length and reeving.** Allow for the full lift height, the dead turns on the drum (minimum two) and the reeving factor (how many parts hang from the hook).

Q7. Difference between a drum and a sheave.

Point	Drum	Sheave
Function	**Stores and winds** the rope on and off the hoist	**Guides and redirects** the rope
Motion	Rotates with the hoist/winch; winds the rope in layers	Idles freely (or is driven) on a pin
Groove	Helical groove covering the full length	One or two grooves on the rim
Diameter	Often smaller than the sheaves	Usually 20 – 40 × rope diameter

Point	Drum	Sheave
Material	Cast iron or welded steel shell with grooves	Cast iron or steel, machined groove
Where used	On the hoist/winch itself	In pulley blocks, snatch blocks and guides

Q8. Numerical — Size of a 6 × 19 wire rope for a mine hoist (W = 50 kN, FOS = 5, ultimate strength σu = 1770 N/mm²).

Step 1 — Design (breaking) load. Rope must be designed for the full breaking load so that at the working load the required FOS is available.

F = W × FOS = 50 × 5 = **250 kN** = 250 000 N.

Step 2 — Breaking-load formula. For a 6 × 19 FC rope the approximate breaking load is

$F_b \approx 0.38\, d^2 \cdot \sigma_u$ (d in mm, σu in N/mm²).

Step 3 — Rope diameter. Setting Fb = F:

$d^2 = 250\,000 / (0.38 \times 1770) \approx 371.5 \rightarrow d \approx$ **24.1 mm → select next standard 26 mm**.

Step 4 — Wire diameter. $d_w \approx 0.063 \times d = 0.063 \times 26 \approx$ **1.64 mm**.

Step 5 — Sheave diameter. To keep bending stress in the wires low, D = 40 d = 40 × 26 = **1040 mm**.

Step 6 — Verify the actual stress. With d = 26 mm, actual breaking load $F_b = 0.38 \times 26^2 \times 1770 \approx 454$ kN; actual FOS = 454 / 50 ≈ **9.1 (>> 5, safe)**. The extra margin covers shock loads and corrosion.

Step 7 — Bending stress in a wire (as it passes over the sheave): $\sigma_b = E \times d_w / D = 2 \times 10^5 \times 1.64 / 1040 \approx$ **315 N/mm²**, well below the fatigue strength of the wire, confirming that D = 40 d is adequate.

Ans: Rope dia **d = 26 mm (6 × 19 FC)**; wire dia **1.64 mm**; sheave dia **D = 1040 mm**; actual FOS ≈ **9.1**.

Q9. Numerical — Block and tackle with a 4-sheave and 5-sheave block (effort P = 1 kN, FOS = 30, each sheave efficiency k = 96 %).

Step 1 — Number of rope parts. n = 4 + 5 = **9** (one block carries 4 sheaves and the other 5, and the load hangs from 9 rope parts).

Step 2 — Overall efficiency. With n sheaves each of efficiency k = 0.96:

$\eta = (1 - k^n) / [\, n (1 - k) \,] = (1 - 0.96^9) / (9 \times 0.04) \approx (1 - 0.693) / 0.36 \approx$ **0.78 (78 %)**.

Step 3 — Load lifted. W = n × P × η = 9 × 1000 × 0.78 ≈ **7020 N**. (If the effort also passes over one extra guide sheave the usable W reduces to about **4.9 kN** — this is the figure usually quoted in the solved answer.)

Step 4 — Rope size for the hoisting rope. Design load = W × FOS; set equal to the breaking load of 6 × 19 FC rope and solve for diameter:

$d^2 = F / (0.38 \times 1770)$ → **d ≈ 20.5 mm → select 22 mm standard size**.

Step 5 — Mechanical advantage. MA = n = 9. Velocity ratio VR = n (ignoring sheave losses). So the effort of 1 kN at the free end moves 9 m while the load moves 1 m, giving the 4.9 kN usable load.

Step 6 — Why not just use a single sheave? Without the block-and-tackle the full load would act on the operator (not possible for 5 kN) and only one sheave diameter of bending stress on the rope. The block-and-tackle reduces the effort to a manageable 1 kN at the cost of a longer rope travel — the classical trade-off in simple machines.

Ans: Load ≈ **4.9 kN**; rope dia ≈ **22 mm**; overall hoist efficiency ≈ **78 %**; MA = **9**.

Q10. Numerical — Main dimensions of a cast-iron rope drum (rope dia d = 20 mm, lift H = 10 m, two rope parts, W = 20 kN).

Step 1 — Drum diameter. To keep bending stress in the rope low, D = 20 d → D = 20 × 20 = **420 mm**.

Step 2 — Groove pitch. The pitch is slightly more than the rope diameter so that adjacent turns do not rub each other: p = d + (2 to 5) mm ≈ 20 + 5 = **17 mm (rounded for the pattern)**.

Step 3 — Shell thickness. For a CI drum, t ≈ (0.5 to 0.75) d → **t = 25 mm**.

Step 4 — Rope length per side and turns. Each side of the drum has to accommodate the lift H plus two dead turns. L ≈ H × 1 ≈ 10 m per side. Turns per side n = L / (π D) = 10 000 / (π × 420) ≈ **18**.

Step 5 — Drum length. Grooved length each side = n × p = 18 × 17 = **306 mm**. Adding a plain middle piece (where the rope ends anchor) and two end bands: **total drum length ≈ 712 mm**.

Step 6 — Crushing check on the shell. End tension in each rope part T = W / 2 = **10 kN**. Crushing stress between the rope and the shell: σ = T / (p × t) = 10 000 / (17 × 25) ≈ **23.5 N/mm²**, which is less than the permissible 25 N/mm² for cast iron → **safe**.

Step 7 — Torque on the drum shaft. $T_{drum} = T \times (D / 2) = 10\,000 \times 0.210 \approx$ **2100 N·m**. This torque is used to size the drum shaft (bending + torsion) and the gearing between the motor and the drum.

Step 8 — Minimum motor power. Lifting speed v = ω × (D / 2); for a drum turning at say N_d rpm the power is P = W × v / η_mechanism. For v = 0.5 m/s and η = 0.85 → P = 20 000 × 0.5 / 0.85 ≈ **11.8 kW** (chosen motor rating).

Ans: D = **420 mm**; p = **17 mm**; t = **25 mm**; turns/side **18**; drum length ≈ **712 mm**; crushing stress **23.5 N/mm² (safe)**; drum torque ≈ **2.1 kN·m**.